Automate Cybersecurity Tasks with Python

Module 1
Introduction to Python

Module 2
Write effective Python code

Module 3
Work with strings and lists

Module 4
Python in practice

Python for automation

- Welcome to module 4
  Video • 1 min
- Automate cybersecurity tasks with Python
  Video • 3 min
- Automating Tasks in CI/CD
  Reading • 10 min
- Essential Python components for automation
  Reading • 8 min
- Clancy: Continual learning and Python
  Video • 2 min
- Test your knowledge: Python and automation
  Practice Assignment • Grade: 100%

Work with files in Python

- Access a text file in Python
  Video • 3 min
- Import files into Python
  Reading • 8 min
- Parse a text file in Python
  Video • 2 min
- Work with files in Python
  Reading • 8 min
- Activity: Import and parse a text file
  Lab • 40 min
- Exemplar: Import and parse a text file
  Lab • 20 min
- Develop a parsing algorithm in Python
  Video • 6 min
- Activity: Create another algorithm
  Lab • 50 min
- Exemplar: Create another algorithm
  Lab • 20 min
- Portfolio Activity: Update a file through a Python algorithm
  Graded Assignment • Grade: 100%
- Portfolio Activity Exemplar: Update a file through a Python algorithm
  Reading • 4 min
- Test your knowledge: Work with files in Python
  Practice Assignment • Grade: 100%

Debug Python code

- Debugging strategies
  Video • 6 min
- Matt: Learning from mistakes
  Video • 3 min
- Apply debugging strategies
  Video • 5 min
- Explore debugging techniques
  Reading • 8 min
- Activity: Debug Python code
  Lab • 35 min
- Exemplar: Debug Python code
  Lab • 20 min
- Test your knowledge: Debug Python code
  Practice Assignment • Grade: 100%

Review: Python in practice

- Coach dialogue: Automate security tasks and parse files with Python
  Dialogue • 15 min
- Wrap-up
  Video • 1 min
- Reference guide: Python concepts from module 4
  Reading • 4 min
- **Glossary terms from module 4**
  Reading • 4 min
- Module 4 challenge
  Graded Assignment • 50 min

Congratulations on completing Course 7!

- Reflect and connect with peers
  Reading • 2 min
- Course wrap-up
  Video • 2 min
- Reference guide: Python concepts from Course 7
  Reading • 4 min
- Course 7 glossary
  Reading • 2 min
- Get started on the next course
  Reading • 4 min

Glossary terms from module 4

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Terms and definitions from Course 7, Module 4

Automation: The use of technology to reduce human and manual effort to perform common and repetitive tasks

Conditional statement: A statement that evaluates code to determine if it meets a specified set of conditions

Debugger: A software tool that helps to locate the source of an error and assess its causes

Debugging: The practice of identifying and fixing errors in code

Exception: An error that involves code that cannot be executed even though it is syntactically correct

File path: The location of a file or directory

Function: A section of code that can be reused in a program

Integrated development environment (IDE): A software application for writing code that provides editing assistance and error correction tools

Iterative statement: Code that repeatedly executes a set of instructions

Log: A record of events that occur within an organization's systems

Logic error: An error that results when the logic used in code produces unintended results

Parsing: The process of converting data into a more readable format

Syntax error: An error that involves invalid usage of a programming language

Variable: A container that stores data

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